INCH-POUND

MIL-T-50333C (AR)
04 December 1991
SUPERSEDING
MIL-T-50333B (AR)
30 August 1984

MILITARY SPECIFICATION

TEST SET, SUBASSEMBLY MX-8048/TPM-22A
(RECEIVER, TEST PANEL)

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for Test Set, Subassembly MX-8048/TPM-22, hereinafter called Receiver Test Panel. This test panel is part of Radar Test Set AN/TPM-22A. It provides means of testing the receiver of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
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MIL-P-50317 - Protector, Over Voltage: 10548031
MIL-R-50318 - Regulator Voltage: 10548034
MIL-R-50319 - Regulator Voltage: 10548043
MIL-T-50320 - Transmit Gate: 10549202
MIL-G-50322 - Generator Digital: 10548613
MIL-P-50324 - Pulse Shaper: 10548801
MIL-C-50344 - Counter, Electronic: 10548009
MIL-S-50348 - Synchronizer, Pulse: 10554894
MIL-G-50349 - Generator, Pulse: 10554892
MIL-G-50351 - Generator: 10558216

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10554527 - Test Set, Subassembly MX-8048/TPM-22
10554597 - Test Set, Subassembly MX-8048/TPM-22 Schematic
10554934 - Test Oscillator

SPECIAL PACKAGING INSTRUCTION

PS10554527 - Packaging of Test Set, Subassembly MX-8048/TPM-22

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however,

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supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10554527, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Receiver Test Panel shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.2.1 Subassemblies. The subassemblies shall have been inspected and accepted in accordance with their respective drawings and specifications prior to their installation in the Receiver Test Panel. The following subassemblies are covered by the indicated specifications:

Unit No.	Name	Specification
1A2A1A2	Transmit Gate: 10549202	MIL-T-50320
1A2A1A4	Generator Digital: 10548613	MIL-G-50322
1A2A1A5	Pulse Shaper: 10548801	MIL-P-50324
1A2A1A7	Counter, Electronic: 10548009	MIL-C-50344
1A2A1A8	Synchronizer, Electrical: 10554894	MIL-S-50348
1A2A1A9	Generator, Pulse: 10554892	MIL-G-50349
1A2A1A10	Regulator Voltage: 10548034	MIL-R-50318
1A2A1A11	Regulator Voltage: 10548043	MIL-R-50319
1A2A1A12	Protector, Over Voltage: 10548031	MIL-P-50317
1A2A1A13	Generator: 10558216	MIL-G-50351

3.3 Environmental service conditions.

3.3.1 Vibration. The Receiver Test Panel shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after

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exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The Receiver Test Panel shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The Receiver Test Panel shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The Receiver Test Panel shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Receiver Test Panel shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32 deg.F and +131 deg.F.

3.3.4.2 Non-operating (storage). The Receiver Test Panel shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg.F for a period of 24 hours and +160 deg.F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The Receiver Test Panel shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the Receiver Test Panel.

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3.5 Performance. The Receiver Test Panel shall operate in two different modes, herein called POWER CONTROL AND SYNC TRIGGER CONTROL.

3.5.1 POWER CONTROL.

3.5.1.1 POWER switch S1. With IF DELAY ADJ switch (COARSE) in MIN position and IF DELAY ADJ switch (FINE) turned fully counterclockwise, and with the inputs specified in Table I (Input characteristics) applied to the Receiver Test Panel, POWER switch S1 shall serve to change indicator DS1 from OFF to ON and the reverse, and to cause the outputs specified in Tables II (Output characteristics) and III (Output characteristics peculiar to power control) to appear at their respective terminals in accordance with POWER indicator DS1.

TABLE I. Input characteristics.

INPUT	CHARACTERISTIC	TERMINALS
Input voltages		
+33 Vdc	+33.0 Vdc ±1.0	J7-A(+), J7-B(RET)
±15 Vdc	±15.0 Vdc ±0.1	J7-E(+), J7-F(RET) J7-G(-)
±23 Vdc	±23.0 Vdc ±1.0	J7-H(+), J7-J(RET) J7-K(-)
+6 Vdc	+6.4 Vdc ±0.1	J7-T(+), J7-U(RET)
+29 Vdc	+29.0 Vdc ±1.0	J7-V(+), J7-W(RET)
+24 Vdc	+24 Vdc +6, -2	J7-S(+), J7-Y(RET)
FRAME GROUND		J7-e
Input signal		
POWER CONTROL	+24 Vdc +6, -2	J7-X(+), J7-Y(RET)

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TABLE II. Output characteristics.

INPUT	CHARACTERISTIC	TERMINAL	TERMINATION
Input voltages			
PRE-RANGE TRIGGER	Pulse train 19,979 ±1 Hz, 0.46 ±0.05 us pulse width, not less than 2.4 V pulse amplitude, delayed not less than 34 us or more than 45 us from MODULATOR TRIGGER.	J2-B	150 ohms
E/L TRIGGER	Pulse train 19,979 ±1 Hz, 0.35 ±0.05 us pulse width, not less than 2.4 V pulse amplitude, delayed not less than 36 us or more than 47 us from Modulator Trigger and delayed not less than 1.7 us from PRE-RANGE TRIGGER.	J2-D	150 ohms
MODULATOR TRIGGER UNIT 3	Pulse train 19,979 ±1 Hz, 0.25 ±0.10 us pulse width, not less than 12 V pulse amplitude, rise time not more than 60 ns at 10 V level.	J2-E	150 ohms (see Note 1)
+15 V 0.5% REG	+15 Vdc (see Note 4)	J3-g	N/A
-15 V 0.5% REG	-15 Vdc (see Note 4)	J3-h	N/A
STALO OSC.	59.936 ±0.001 MHz (CW) using AR1 adjust voltage amplitude of 6.7 +0.4/-0.3 V P-P, into a 50 ohm load.	J4-A	50 ohms
VIDEO PULSE 3A2	Pulse train 19,979 ±1 Hz, 0.2 to 0.5 us pulse width at the 50% points, delayed not greater than 2.0 us from MODULATOR TRIGGER. This signal shall vary in amplitude with changes in the position of VIDEO ADJUST control R1, from 0 volts when control is turned fully counterclockwise, to not less than 2.4 V when control is turned fully clockwise.	J8	150 ohms

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TABLE II. Output characteristics (continued).

OUTPUT	CHARACTERISTIC	TERMINAL	TERMINATION
3A2/3A5 TRIGGER	Same as PRE RANGE TRIGGER (see above) except amplitude shall vary with changes in position of 3A2/3A5 TRIGGER ADJUST control R2, from 0 V when control is turned fully counterclockwise, to not less than 2.4 V when control is turned fully clockwise.	J10	150 ohms
LO OUT (1)	59.936 ±0.001 MHz, 1.10 ± 0.25 V P-P amplitude, into 50 ohm load.	J13	50 ohms
LO OUT (2)	Same as LO OUT (1) (see above).	J14	50 ohms
LO 3A5(1)	59.936 ±0.001 MHz, 3.44 ± 0.56 V P-P amplitude, into 50 ohm load.	J20	50 ohms
LO 3A5(2)	Same as LO 3A5(1) (see above).	J9	50 ohms
3A9 TRIGGER	Same as MODULATOR TRIGGER UNIT 3 (see above).	J21 (see Note 2)	150 ohms
MODULATOR TRIGGER	Pulse train 19,979 ±1 Hz, pulse width of 1.0 ±0.4 us, pulse amplitude of 20 ±5 V.	J22	150 ohms
SYNC TRIGGER	Pulse train 19,979 ±1 Hz, Pulse width of 1.0 ±0.4 us, pulse amplitude of 20 ±5 V, coincident with MODULATOR TRIGGER.	J24	150 ohms
IF DELAY TRIGGER PULSE	Pulse train 20 ±1% KHz, pulse width of 250 ± 100 nsec amplitude of not less than +12 V and a rise time of less than 60 ns.	J34	50 ohms

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TABLE II. Output characteristics (continued).

NOTES:

1. When observing the MODULAR TRIGGER display at J2E with a 150 ohm termination, remove temporarily the 150 ohm termination at J21.

2. When observing the 3A9 TRIGGER output display at J21 with the 150 ohm termination, remove temporarily the 150 ohm termination at J2-E.

3. Output pulse amplitude and their width tolerances may change by ±10% at temperature extremes.

4. With power pushbutton on, 15 volt within .050 V of respective 15 V input, Table I.

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TABLE III. Output characteristics peculiar to power control.

OUTPUT	CHARACTERISTIC	TERMINAL	TERMINATION
+25 V REG	25 Vdc ± 0.5 Vdc	J3-M(+), J3-N(RET)	N/A
-15 V REG	-15 Vdc ±0.100*	J3-a(-), J3-m(RET)	N/A
+15 V REG	+15 Vdc ± 0.100*	J3-b(+), J3-m(RET)	N/A

NOTE:

* Tolerance permitted to degrade to ±0.300 V at temperature extremes.

3.5.1.2 CLUTTER LOCK ON switch S2. When the Receiver Test Panel is supplied with the +24 Vdc input voltage and the POWER CONTROL input signal of Table I, and with POWER switch S1 turned ON, CLUTTER LOCK ON switch S2 shall serve to illuminate and extinguish CLUTTER LOCK ON indicator DS2 and the reverse, and to apply 24 Vdc +6, -2 signal to terminal J3-W in accordance with indicator DS2. It shall also perform functions discussed in paragraph 3.5.1.6.

3.5.1.3 MID RANGE CALIBRATION switch S3. With IF DELAY ADJ switch (COARSE) in MIN position, IF DELAY ADJ control (FINE) turned fully counterclockwise, the inputs specified in Table I applied to the Receiver Test Panel, the POWER switch S1 turned ON, MID RANGE CALIBRATION switch S3 shall serve to illuminate and extinguish MID RANGE CALIBRATION indicator DS3 and the reverse, to cause the outputs of Table IV (Output characteristics peculiar to mid range calibration) to appear at their respective terminals in accordance with indicator DS3, and to perform functions discussed in paragraph 3.5.1.6.

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TABLE IV. Output characteristics peculiar to mid range calibration.

OUTPUT	CHARACTERISTIC	TERMINALS	TERMINATION
MID RANGE CALIBRATION CONTROL	With MID RANGE CALIBRATION indicator DS3 illuminated, +24 Vdc +6, -2.	J3-A(+) J3-T(RET)	N/A
RANGE CALIBRATION RF INPUT	(1) With MID RANGE CALIBRATION indicator DS3 illuminated, 59.936 +0.001 MHz burst, 0.28 ±0.1 us wide, measured at the 50% point, not less than 80 mV P-P amplitude into 50 ohm load, delayed not greater than 2.0 us from MODULATOR TRIGGER. The signal shall remain when S3 is extinguished. (2) With MID RANGE CALIBRATION indicator DS3 extinguished, ground.	J4-G	50 ohms

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3.5.1.4 RECEIVER NOISE switch S4. When the Receiver Test Panel is supplied with the +24 Vdc input voltage and the POWER CONTROL Input signal of Table I, and with POWER switch S1 turned ON, RECEIVER NOISE switch S4 shall serve to illuminate and extinguish RECEIVER NOISE indicator DS4 and the reverse. It shall also perform functions discussed in paragraph 3.5.1.6.

3.5.1.5 3A6 TEST switch S5. When the Receiver Test Panel is supplied with the +24 Vdc input voltage and the POWER CONTROL input signal of Table I, and the POWER switch S1 turned ON, 3A6 TEST switch S5 shall serve to illuminate and extinguish 3A6 TEST indicator DS5 and the reverse. It shall also perform functions discussed in paragraph 3.5.1.6.

3.5.1.6 Combinations of switches S1, S2, S3, S4 and S5. With IF DELAY ADJ switch (coarse) in MIN position, IF DELAY ADJ control (fine) turned fully counterclockwise, IF AMPLITUDE ADJ control AT4 turned to the zero (00) position, the inputs specified in Table I applied to the Receiver Test Panel and switches S1, S2, S3, S4 and S5 shall serve to cause the outputs of Table V (Outputs for combinations of indicators DS1, DS2, DS3, DS4 and DS5) to appear at their respective terminals in accordance with the combinations of indicators DS1, DS2, DS3, DS4 and DS5 shown in Table VI (Combinations of indicators DS1, DS2, DS3, DS4 and DS5). Prior to starting the A through P combinations of indicators as shown in Table V, satisfy Combination Q, at J25 of the receiver test panel into 50 ohms, by first applying +20 to +30 Vdc to pin J7-P with S1 through S5 in the off position and J7 connected. Next, measure the burst width to verify the requirements of Table VII (Output characteristics peculiar to sync trigger control) GATED STALO OUTPUT and then adjust AR2 for a burst amplitude of 1.2 V min to 1.5 V P-P max.

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TABLE V. Outputs for combinations of indicators DS1, DS2, DS3, DS4 and DS5.

OUTPUT	CHARACTERISTICS	INDICATOR COMBINATION (TABLE VI)	TERMINAL	TERMINATION
IF INPUT UNIT 3 Measured at 50% Points	59.936 ± 0.001 MHz burst, 0.28 ± 0.02 us wide, amplitude of 0.950 to 1.500 V P-P into 50 ohm load delayed not greater than 2.0 us from MODULATOR TRIGGER.	F, L, M, P	J1-D	50 ohms
	59.941 ± 0.001 MHz burst 0.28 ± 0.02 us wide, amplitude of 0.950 to 1.500 V P-P into 50 ohm load delayed not greater than 2.0 us from MODULATOR TRIGGER.	A, D, E, K		50 ohms
	59.936 ± 0.001 MHz burst superimposed on a 59.941 ± 0.001 MHz CW. The burst shall be 0.28 ± 0.02 us wide, 0.950 to 1.250 V P-P amplitude into a 50 ohm load, delayed not greater than 2.0 us from MODULATOR TRIGGER. The CW shall have an amplitude of 0.556 to 1.65 V peak-to-peak.	C, H, J, O		50 ohms
IF OUT Measured at 50% Points	59.936 ± 0.001 MHz burst superimposed on a 59.941 ± 0.001 MHz CW. The burst shall be 0.28 ± 0.02 us wide, 54 to 105 mV P-P amplitude into a 50 ohm load, delayed not greater than 2.0 us from MODULATOR TRIGGER. The CW shall have an amplitude of 43 to 80 mV peak-to-peak.	C, H, J, O	J11	50 ohms

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TABLE V. Outputs for combinations of indicators DS1, DS2, DS3, DS4 and DS5 (continued).

OUTPUT	CHARACTERISTICS	INDICATOR COMBINATION (TABLE V)	TERMINAL	TERMINATION
	59.936 ± 0.001 MHz burst 0.28 ± 0.02 us wide, amplitude of 54 to 105 mV P-P into 50 ohm load delayed not greater than 2.0 us from MODULATOR TRIGGER.	F, L, M, P		50 ohms
	59.941 ± 0.001 MHz burst 0.28 ± 0.02 us wide, amplitude of 54 to 126 mV P-P into 50 ohm load, delayed not greater than 2.0 us from MODULATOR TRIGGER.	A, D, E, K		50 ohms
(Remove DC3-J3 to measure burst amplitude. Return connections after measurements.)		Remove DC3-J2 to measure CW amplitude.		
IF OUT	Same as IF OUT	Same as IF OUT (J11)	J12	50 ohms

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TABLE VI. Combinations of indicators DS1, DS2, DS3, DS4 and DS5.

NOTE: DS1 legend is ON/OFF.
DS2 legend is CLUTTER LOCK ON.
DS3 legend is MID RANGE CALIBRATION.
DS4 legend is RECEIVER NOISE.
DS5 legend is 3A6 TEST.

In the table below, "on" means legend is illuminated, or bright, and "off" means legend is extinguished or dark.

INDICATOR COMBINATIONS	COMBINATION LETTER CODE
DS1 on, DS2 on, DS3 on, DS4 on, DS5 on	A
DS1 on, DS2 on, DS3 on, DS4 on, DS5 off	B
DS1 on, DS2 on, DS3 on, DS4 off, DS5 on	C
DS1 on, DS2 on, DS3 off, DS4 on, DS5 on	D
DS1 on, DS2 off, DS3 on, DS4 on, DS5 on	E
DS1 on, DS2 on, DS3 on, DS4 off, DS5 off	F
DS1 on, DS2 on, DS3 off, DS4 on, DS5 off	G
DS1 on, DS2 on, DS3 off, DS4 off, DS5 on	H
DS1 on, DS2 off, DS3 on, DS4 on, DS5 off	I
DS1 on, DS2 off, DS3 on, DS4 off, DS5 on	J
DS1 on, DS2 off, DS3 off, DS4 on, DS5 on	K
DS1 on, DS2 on, DS3 off, DS4 off, DS5 off	L
DS1 on, DS2 off, DS3 on, DS4 off, DS5 off	M
DS1 on, DS2 off, DS3 off, DS4 on, DS5 off	N

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TABLE VI. Combinations of indicators DS1, DS2, DS3, DS4 and DS5 (continued).

INDICATOR COMBINATIONS	COMBINATION LETTER CODE
DS1 on, DS2 off, DS3 off, DS4 off, DS5 on	O
DS1 on, DS2 off, DS3 off, DS4 off, DS5 off	P
DS1 off, DS2 off, DS3 off, DS4 off, DS5 off	Q*

*Perform this combination first to adjust AR2 (see paragraph 3.5.1.6).

TABLE VII. Output characteristics peculiar to sync trigger control.

OUTPUT	CHARACTERISTIC	TERMINAL	TERMINATION
GATED STALO OUTPUT Measured at 50% points	59.936 ±0.001 MHz burst 260 to 300 ns wide, with AR2 adjust to obtain an amplitude of 1.2 to 1.5 V P-P into a 50 ohm load, delayed not greater than 2.0 us from MODULATOR TRIGGER. (See Indicator of Combinations Q of Table VI. Perform adjustment Q prior to any other combinations.)	J25	50 ohms
STALO OSC UNIT II	59.936 ±0.001 MHz, 6.7 +0.4 -0.3 V P-P amplitude into 50 ohm load.	J23	50 ohms

3.5.2 SYNC TRIGGER CONTROL. With IF DELAY ADJ switch (COARSE) in MIN position, and IF DELAY ADJ control (FINE) turned fully counterlockwise, and when supplied with the inputs specified in Table VIII (Input characteristics) the receiver test panel shall yield the outputs specified in Tables II and VII.

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TABLE VIII. Input characteristics.

INPUT	CHARACTERISTIC	TERMINALS
Input voltages		
+33 Vdc	+33.0 Vdc ±1.0	J7-A(+), J7-B(RET)
±15 Vdc	±15.0 Vdc ±0.1	J7-E(+), J7-F(RET) J7-G(-)
+6 Vdc	6.0 Vdc +0, -0.4	J7-T(+), J7-U(RET)
+29 Vdc	+29.0 Vdc ±1.0	J7-V(+), J7-W(RET)
+24 Vdc	+24 Vdc +6, -2	J7-S(+), J7-Y(RET)
FRAME GROUND		J7-e
Input signal		
SYNC TRIGGER CONTROL	+24 Vdc +6, -2	J7-P(+), J7-Y(RET)

3.5.3 IF DELAY ADJ controls. The IF DELAY ADJ controls (COARSE and FINE) shall serve to change the time delay of the following signals relative to the MODULATOR TRIGGER signal in accordance with Table IX (IF delay adjustments).

Signal	Table in Which Described
GATED STALO OUTPUT	VII
RANGE CAL RF INPUT	IV
IF INPUT UNIT 3	V
IF OUT	V
IF OUT 3A6	V

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TABLE IX. IF delay adjustments.

POSITION OF COARSE CONTROL	POSITION OF FINE CONTROL	DELAY RELATIVE TO MODULATOR TRIGGER
MIN	Fully CCW	Delay not greater than 2.0 us
	INTERMEDIATE	Delay not greater than 9.0 us
	Fully CW	Delay not less than 9.0 us
PHASE	Fully CCW	Delay not greater than 39.0 us
	MID RANGE	Delay more than 35.0 us and less than 44.0 us
	Fully CW	Delay not less than 46.0 us
MAX	Fully CCW	Delay not greater than 49.0 us
	MID RANGE	Delay not more than 45.0 us and less than 54.0 us
	Fully CW	Delay not less than 56.0 us

The IF DELAY ADJ controls (COARSE and FINE) shall serve to change the time delay of the VIDEO PULSE 3A2 signal of Table II relative to the MODULATOR TRIGGER signal in accordance with Table IX, except that every delay of Table IX (other than the first two figures given) shall be reduced by approximately 1.4 us.

3.5.4 IF AMPLITUDE ADJ control. The IF AMPLITUDE ADJ control shall serve to change the amplitude of the signals of Table V in accordance with Table X (IF amplitude adjustment).

TABLE X. IF amplitude adjustment.

SIGNAL	SETTING OF IF AMPLITUDE ADJ	AMPLITUDE OF SIGNAL
IF INPUT UNIT 3	00	As stated in Table V.
	01 to 99	For the F, L, M, P indicator combinations, the 15 DB setting shall yield an amplitude of 208 ±68 mv P-P.

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TABLE X. IF amplitude adjustment. (continued)

SIGNAL	SETTING OF IF AMPLITUDE ADJ	AMPLITUDE OF SIGNAL
IF OUT	00	As stated in Table V.
	01 to 99	For the F, L, M, P indicator combinations, the 10 DB setting shall yield an amplitude of 29.0 ±13.0 mV P-P.
IF OUT 3A6	00 to 99	Same as IF OUT above.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50° to +90°F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) receiver test panels for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table XI (Functional tests) and Table XII (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted

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for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table XI shall be performed on a 100% basis.

TABLE XI. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

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4.5 Special sampling. One Receiver Test Panel out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The same shall have met the requirements and tests of Table XI and shall then meet the tests in Table XII. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XII. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in

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the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XIII (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE XIII. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power supply	33 ± 1 Vdc, current capacity 0.5 A, ripple voltage not greater than 250 mV rms.
2.	Power supply	15.0 ± 0.1 Vdc, current capacity 0.5 A, ripple voltage not greater than 10 mV rms.
3.	Power supply	-15.0 ± 0.1 Vdc, current capacity 0.5 A, ripple voltage not greater than 10 mV rms.
4.	Power supply	+6.4 ± 0.1 Vdc, current capacity 0.5 A, ripple voltage not greater than 10 mV rms.
5.	Power supply	+29.0 ± 0.1 Vdc, current capacity 0.5 A, ripple voltage not greater than 250 mV rms.
6.	Power supply	+26.0 ± 4.0 Vdc, current capacity 1.0 A, ripple voltage not greater than 250 mV rms.
7.	Power supply	23.0 ± 1.0 Vdc, current capacity 0.5 A, ripple voltage not greater than 100 mV rms.
8.	Power supply	-23.0 ± 1.0 Vdc, current capacity 0.5 A, ripple voltage not greater than 100 mV rms.
9.	Standard measuring equipment (SME)	To indicate continuity from 0 to infinity.

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10.	Oscilloscope	Dual channel. Vertical passband DC to 150 MHz. Rise time of scope plus measuring probe 3.5 nanoseconds or less. Sweep period 50 nanoseconds per cm or less. Input resistance greater than 1 megohm. Input capacitance less than 25 picofarads.
11.	Oscilloscope	Dual channel. Vertical passband DC to 15 MHz. Rise time of scope plus measuring probe 27 nanoseconds or less. Sweep period 100 nanoseconds per cm or less. Input resistance greater than 1 megohm. Input capacitance less than 25 picofarads.
12.	Electronic counter	Input sensitivity at least 200 mV rms. Registration of six places or more. Range 10.1 to 100 MHz. Input impedance 1 megohm or greater.
13.	DC digital voltmeter	Visual display 4 digits plus one over-range. Full scale range 100.00 volts or greater. Input resistance 10 megohms or greater. Response time 250 ms or less. Accuracy $\pm$0.01% of reading and $\pm$0.01% of full scale reading.
14.	TPS "T"	Connector.
15.	TPS to BNC	Adapter.
16.	Test harnesses	Per Figures 1 and 2 and Table XIV.

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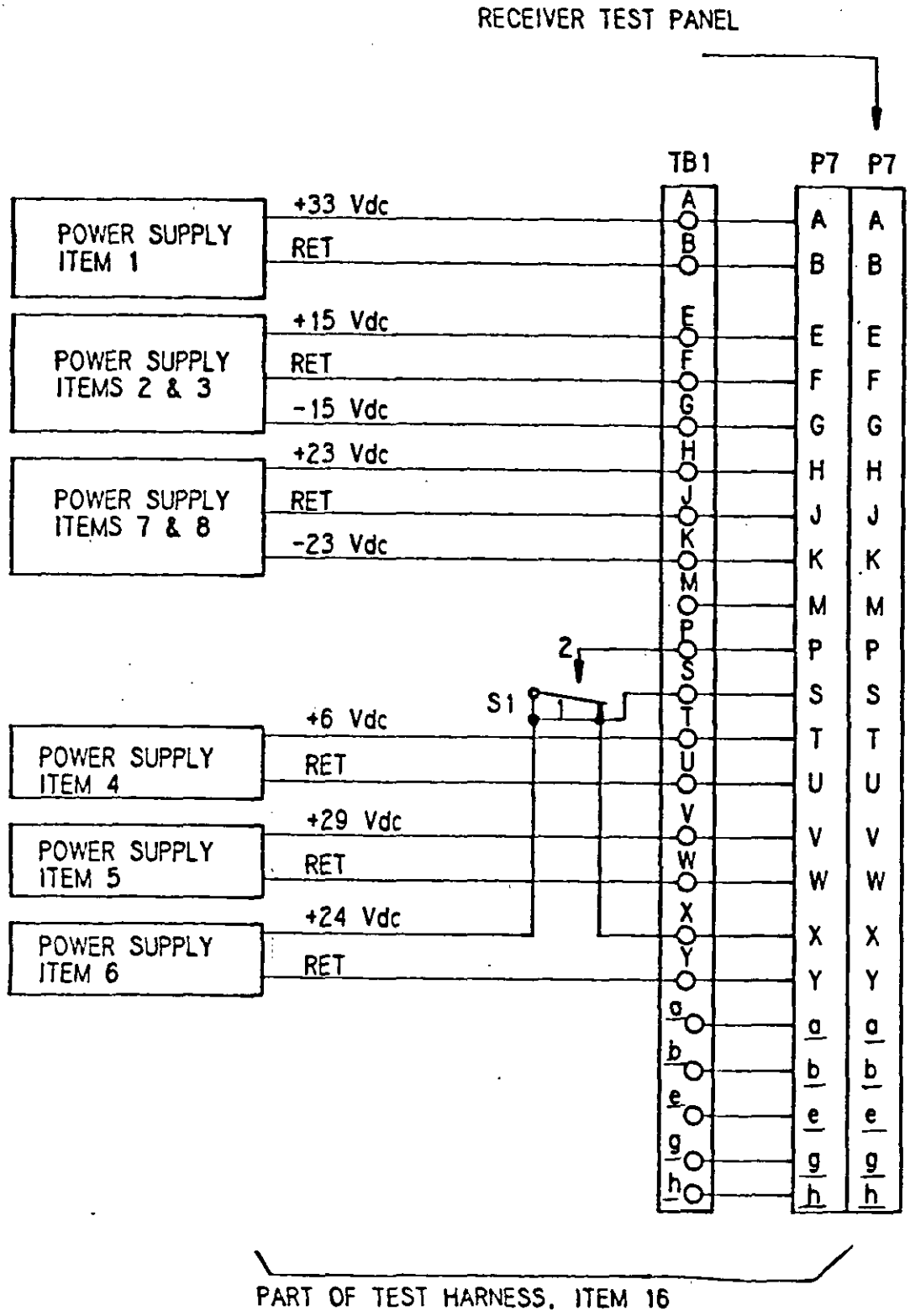


FIGURE 1. Power supplies and harness.

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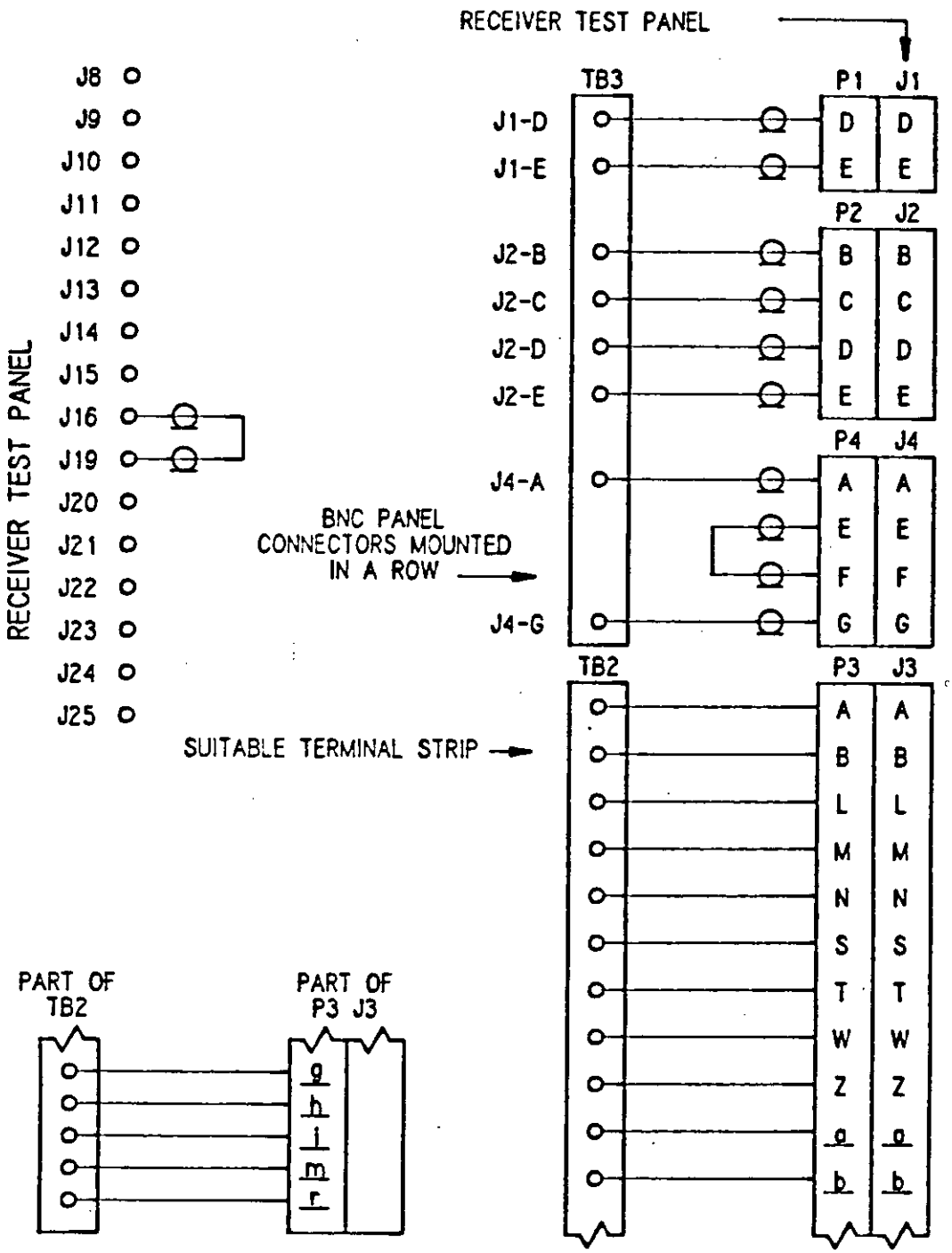


FIGURE 2. Test harness.

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TABLE XIV. Parts list for test harness, item 16.

DESIGNATION	PART	DESCRIPTION
P1	Connector	PT06SE-22-78SX, Bendix, Sidney, NY, or equivalent.
P2	Connector	PT06SE-22-78SW, Bendix, Sidney, NY, or equivalent.
P3	Connector	MS3116E-20-39P (SR).
P4	Connector	PT06SE-22-78S, Bendix, Sidney, NY, or equivalent.
P5	Connector	PT06SE-18-30S, Bendix, Sidney, NY; DTK06-18-30S, Deutsch, Banning, CA, or equivalent.
S1	Switch	SPDT

BNC connectors, cables, terminal strips and wiring per Figures 1 and 2.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The receiver test panel shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the Receiver Test Panel shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

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d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the receiver test panel shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4 and the following:

Connect test harnesses (item 16, Table XIII) to receiver test panel as shown in Figures 1 and 2. Use standard measuring equipment (item 9, Table XIII) to determine the continuity (less than 2 ohms) exists between the points shown in Table XV (Continuity).

TABLE XV. Continuity.

FROM TEST HARNESS TERMINAL	TO TEST HARNESS TERMINAL
TB1-J7-e	TB2-J3-B, L, S, Z, i
TB1-J7-Y	TB3-J3-T
TB-J7-F	TB2-J3-r
TB-J7-B	TB2-J3-N

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Adjust power supplies (items 1 through 8, Table XIII) to deliver outputs described in Tables I and VIII. Connect power supplies to test harness in accordance with Figure 1 so as to apply the input voltages and signal of Table I to the Receiver Test Panel. Place switch S1 of Figure 1 in position 1. Connect test equipment as in Figure 3. Adjust the A channel of oscilloscope to display STALO OSC signal of Table II. Check that signal on channel B is essentially the same as that on channel A.

4.7.3 Performance. Adjust power supplies (items 1 through 8, Table XIII) to deliver outputs described in Tables I and VIII. Attach test harnesses (item 16, Table XIII) to Receiver Test Panel as shown in Figures 1 and 2. Terminate TB3 test harness terminals J1-D, J4-A and J4-G with 50 ohms, and terminate the following receiver test panel terminals with 50 ohms: J9, J11, J12, J13, J14, J20, J23 and J25. Terminate the following Receiver Test Panel terminals with 150 ohms: J2-B, J2-D, J8, J10, J21, J22, J24 and J2-E.

4.7.3.1 POWER CONTROL. Connect power supplies to test harness in accordance with Figure 1 so as to apply the input voltages and signal of Table I to the Receiver Test Panel. Place switch S1 of Figure 1 in position 1. Turn IF DELAY ADJ switch (COARSE) to MIN position. Turn IF DELAY ADJ control (FINE) fully counterclockwise.

4.7.3.1.1 POWER switch S1. Either the OFF or ON legend of POWER indicator DS1 shall be illuminated. Depress POWER switch S1 three times and verify that it causes OFF legend to be extinguished and ON legend to be illuminated, and the reverse. Leave POWER switch S1 ON.

a. ODULATOR TRIGGER. Connect test equipment as in Figure 4. Verify that the amplitude and width of the pulses conform to the requirements of MODULATOR TRIGGER, Table II. Verify that signal disappears when POWER switch S1 is OFF. Connect test equipment as in Figure 5. Adjust electronic counter (item 12, Table XIII) to verify that frequency of MODULATOR TRIGGER is 19,979 ±1 Hz. Return to the setup of Figure 4.

b. PRE RANGE TRIGGER. Remove 150 ohm termination from test harness terminal TB3-J2-B. Connect this terminal to coaxial lead Z of Figure 4. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width and delay of the pulses on channel B conform to the requirements of PRE RANGE TRIGGER, Table II. Remove coaxial lead Z from TB3-J2-B and restore 150 ohm termination at that point.

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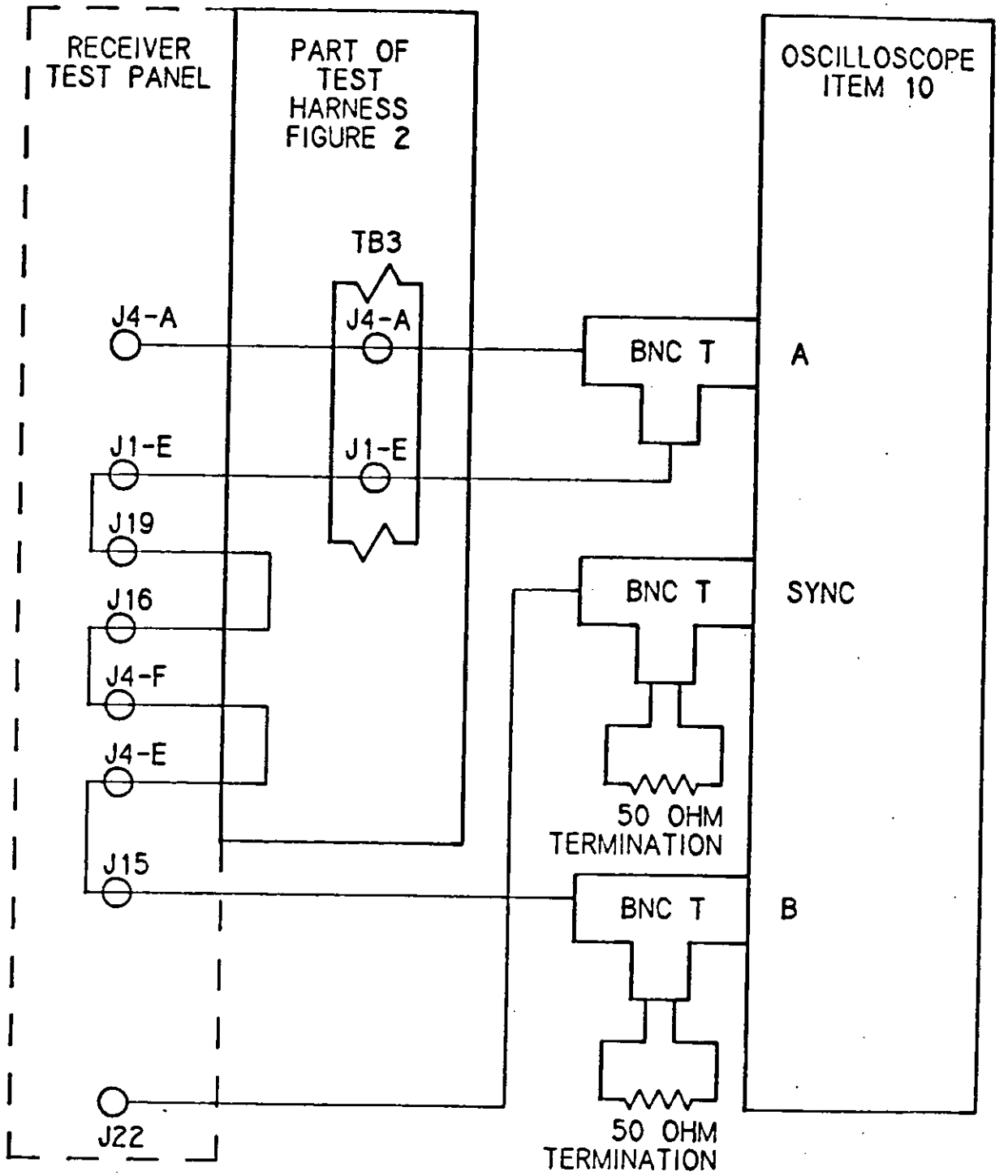


FIGURE 3. Coaxial continuity and high frequency check.

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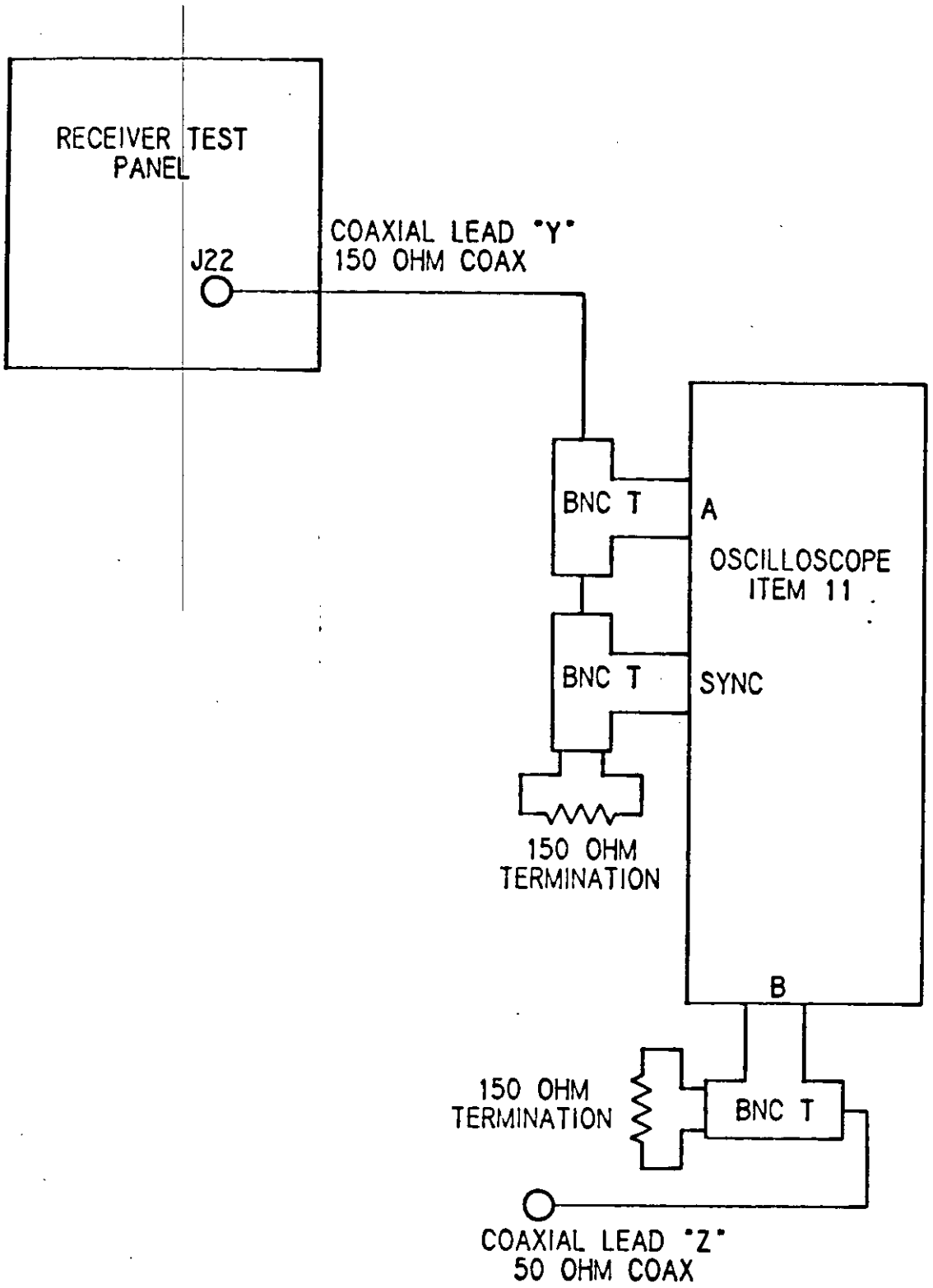


FIGURE 4. Modulator trigger check.

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c. E/L TRIGGER. Remove 150 ohm termination from test harness terminal TB3-J2-D. Connect this terminal to coaxial lead Z of Figure 4. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that amplitude, width and delay of the pulses on channel B conform to the requirements of E/L TRIGGER, Table II. Remove coaxial lead Z from TB3-J2-D and restore 150 ohm termination at that point.

d. MODULATOR TRIGGER Unit 3. Remove 150 ohm termination from test harness terminal TB3-J2-E. Connect this terminal to coaxial lead Z of Figure 4. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude and width of the pulses on channel B conform to the requirements of MODULATOR TRIGGER, Unit 3, Table II. Remove coaxial lead Z from TB3-J2-E and restore 150 ohm termination at that point.

e. VIDEO PULSE 3A2. Remove 150 ohm termination from Receiver Test Panel terminal J8. Connect this terminal to coaxial lead Z of Figure 4. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width and delay of the pulses on channel B conform to the requirements of VIDEO PULSE 3A2, Table II. Verify that VIDEO ADJUST control R1 affects pulse amplitude as specified in VIDEO PULSE 3A2, Table II. Remove coaxial lead Z from J8 and restore 150 ohm termination at that point.

f. 3A2/3A5 TRIGGER. Remove 150 ohm termination from Receiver Test Panel terminal J10. Connect this terminal to coaxial lead Z of Figure 4. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width and delay of the pulses on channel B conform to the requirements of 3A2/3A5 TRIGGER, Table II. Verify that TRIGGER ADJUST control R2 affects pulse amplitude as specified in 3A2/3A5 TRIGGER, Table II. Remove coaxial lead Z from J10 and restore 150 ohm termination at that point.

g. 3A9 TRIGGER. Remove 150 ohm termination from Receiver Test Panel terminal J21. Connect this terminal to coaxial lead Z of Figure 4. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width and delay of the pulses on channel B conform to the requirements of 3A9 TRIGGER, Table II. Remove coaxial lead Z from J21 and restore 150 ohm termination at that point.

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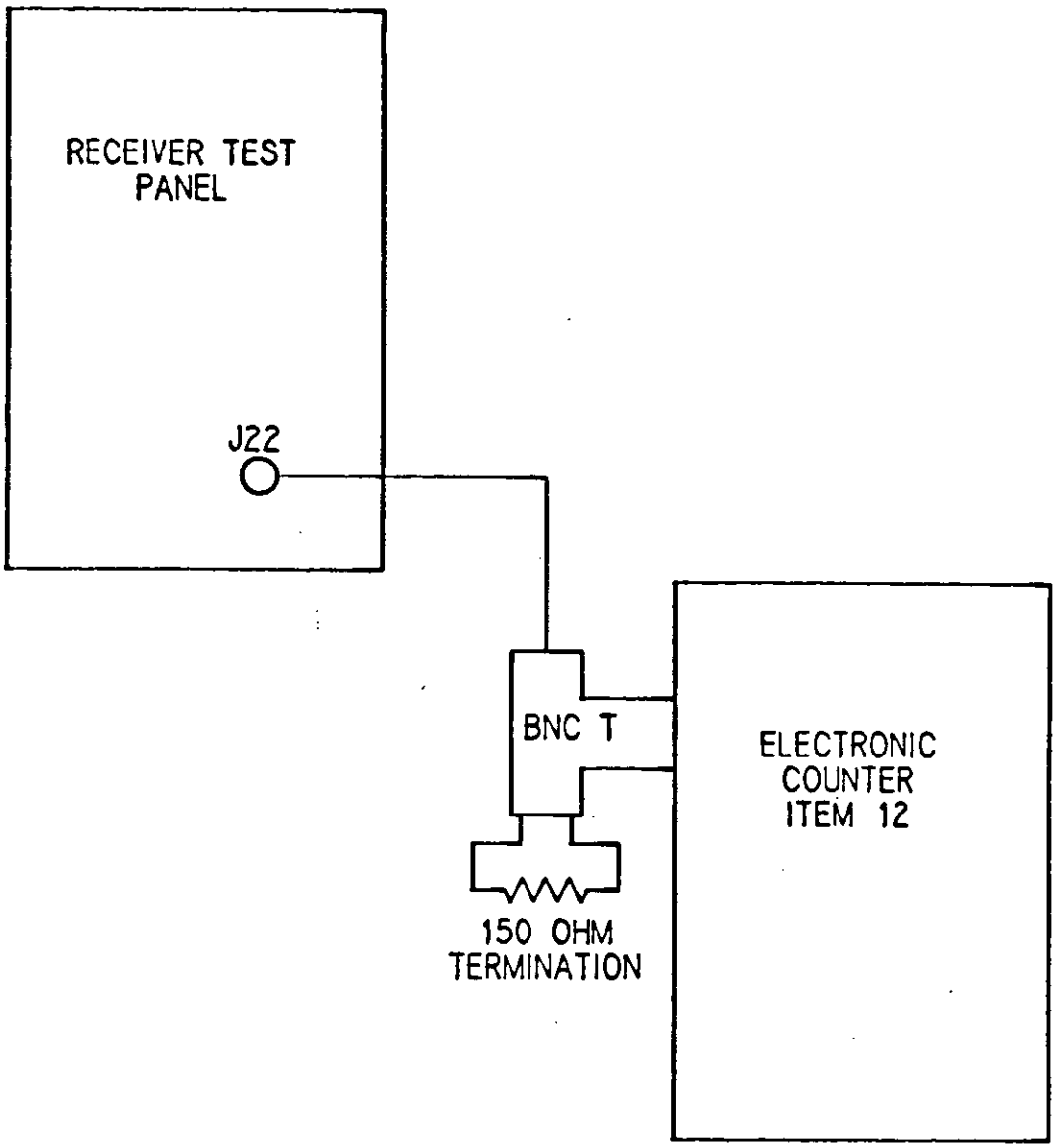


FIGURE 5. Modulator trigger frequency check.

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h. SYNC TRIGGER. Remove 150 ohm termination from Receiver Test Panel terminal J24. Connect this terminal to coaxial lead Z of Figure 4. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width and delay of the pulses on channel B conform to the requirements of SYNC TRIGGER, Table II. Remove coaxial lead Z from J24 and restore 150 ohm termination at that point.

i. +15 V 0.5% REG. Use digital voltmeter (item 13, Table XIII) to verify that voltage between test harness terminals TB2-J3-g(+) and TB2-J3-r conforms to the requirement of +15 V 0.5% REG, Table II.

j. -15 V 0.5% REG. Use digital voltmeter (item 13, Table XIII) to verify that voltage between test harness terminals TB2-J3-h and TB2-J3-r conforms to the requirements of -15 V 0.5% REG, Table II.

k. STALO OSC. Connect test equipment as in Figure 6. Adjust signal amplitude in accordance with Table II STALO OSC using AR1. After adjustment verify that amplitude of signal is constant and that it conforms to the requirements of STALO OSC, Table II. Reconnect test equipment as in Figure 5, except with coaxial lead connected to test harness terminal TB3-J4-A. Verify that frequency of signal conforms to requirement of STALO OSC, Table II. Remove coaxial lead from TB3-J4-A and restore 50 ohm termination at that point.

l. LO OUT (1). Remove 50 ohm termination from Receiver Test Panel terminal J13. Connect test equipment as in Figure 7. Place 3A6 TEST switch (S5) "ON" and receiver noise switch (S4) "OFF." Verify that amplitude of signal is constant and that it conforms to the requirement of LO OUT (1), Table II. Reconnect test equipment as in Figure 5, except with coaxial lead connected to Receiver Test Panel terminal J13. Verify that frequency of signal conforms to requirement of LO OUT (1), Table II. Remove coaxial lead from J13 and restore 50 ohm termination at that point.

m. LO OUT (2). Repeat step l above, except substitute J14 for J13 throughout and LO OUT (2) for LO OUT (1).

n. LO 3A5 (1). Repeat step l above, except substitute J20 for J13 throughout, and LO 3A5 (1) for LO OUT (1).

o. LO 3A5 (2). Repeat step l above, except substitute J9 for J13 throughout, and LO 3A5 (2) for LO OUT (1).

p. +25 V REG. Use digital voltmeter (item 13, Table XIII) to verify that voltage between test harness terminals TB2-J3-M(+) and TB2-J3-N(RET) conforms to the requirement of +25 V REG, Table III. Verify that when OFF legend of POWER indicator DS1 is illuminated, +25 V REG does not appear at TB2-J3-M.

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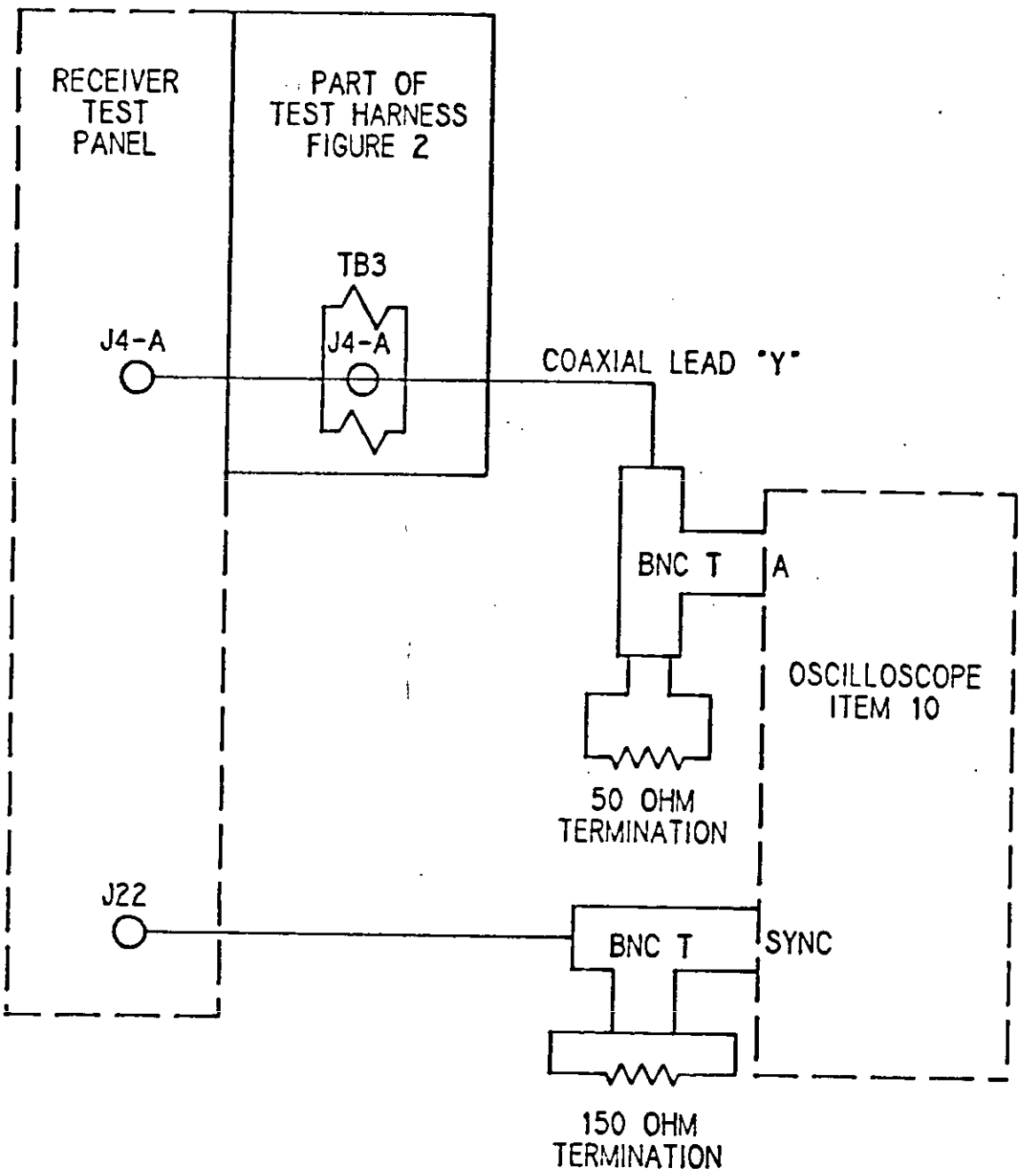


FIGURE 6. Stalo OSC check.

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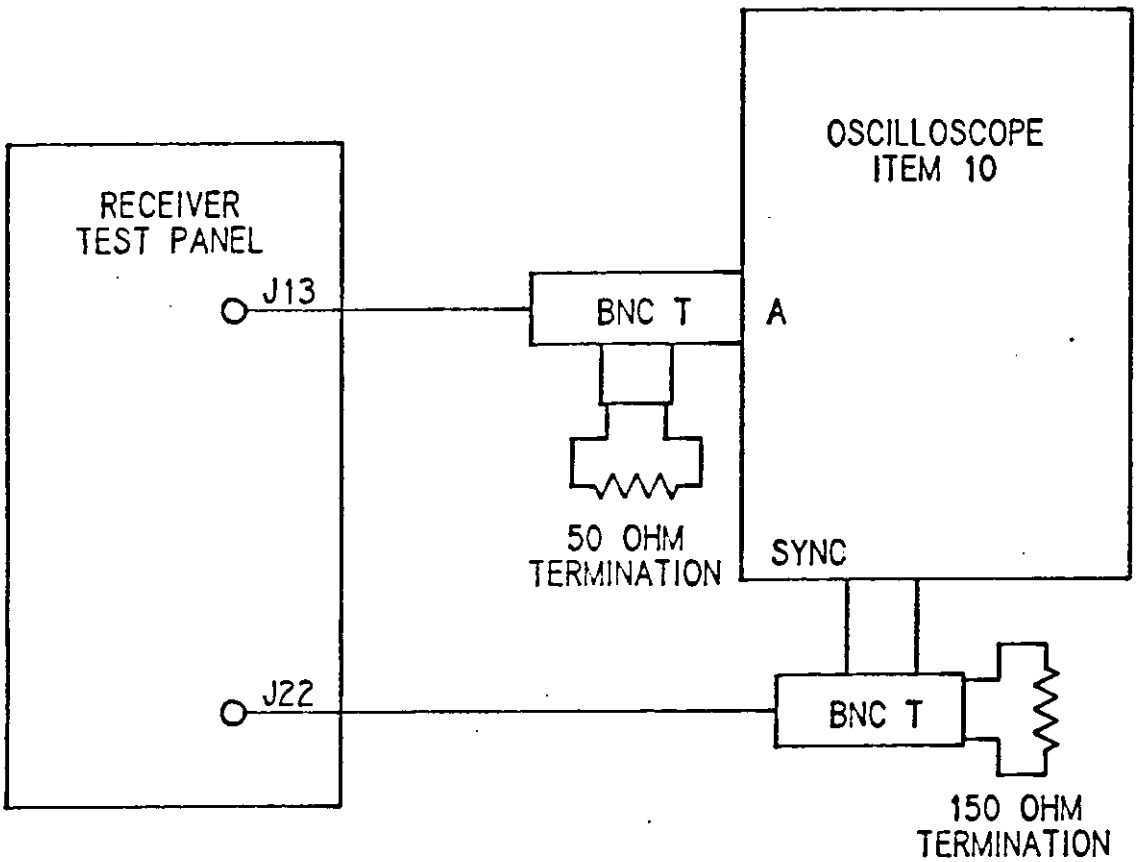


FIGURE 7. Lo out check.

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q. -15 V REG. Use digital voltmeter (item 13, Table XIII) to verify that voltage between test harness terminals TB2-J3-a(-) and TB2-J3-m(RET) conforms to the requirement of -15 V REG, Table III. Verify that when OFF legend of POWER indicator DS1 is illuminated, -15 V REG does not appear at TB1-J3-a.

r. +15 V REG. Use digital voltmeter (item 13, Table XIII) to verify that voltage between test harness terminals RB2-J3-B(+) and TB2-J3-m(RET) conforms to the requirement of +15 V REG, Table III. Verify that when OFF legend of POWER indicator DS1 is illuminated, +15 V REG does not appear at TB2-J3-b. Leave ON legend of POWER indicator DS1 illuminated.

s. IF DELAY TRIG PULSE. Connect the Receiver Test Panel terminal J34 to coaxial lead Z of Figure 4, except substitute 50 ohm terminal for 150 ohm termination. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width and rise time of the pulses on channel B conform to the requirements of IF DELAY TRIG PULSE, Table II.

4.7.3.1.2 CLUTTER LOCK ON switch S2. Depress CLUTTER LOCK ON switch S2 three times and verify CLUTTER LOCK ON indicator DS2 alternately is illuminated and extinguished. Use digital voltmeter (item 13, Table XIII) to verify that +24 +6/ -2 Vdc signal appears between test harness terminals TB2-J3-W(+) and TB2-J3-T(RET) when CLUTTER LOCK ON indicator DS2 is illuminated.

4.7.3.1.3 MID RANGE CALIBRATION switch S3. Depress MID RANGE CALIBRATION switch S3 three times and verify that MID RANGE CALIBRATION indicator DS3 alternately is illuminated and extinguished.

a. MID RANGE CALIBRATION CONTROL. Use digital voltmeter (item 13, Table XIII) to verify that voltage between test harness terminal TB2-J3-A(+) and TB2-J3-T(RET) conforms to the requirement of MID RANGE CALIBRATION CONTROL, Table IV.

b. RANGE CAL RF INPUT. Connect test equipment as in Figure 8. Remove 50 ohm termination from test harness terminal TB3-J4-G and attach coaxial lead Z to this terminal. With indicator DS3 illuminated, verify that the width, amplitude and delay of the bursts on channel B conform to the requirements of RANGE CAL RF INPUT, part 1, Table IV. Remove coaxial lead Z from TB3-J4-G and restore 50 ohm termination.

4.7.3.1.4 RECEIVER NOISE switch S4. Depress RECEIVER NOISE switch S4 three times and verify that RECEIVER NOISE indicator DS4 is alternately illuminated and extinguished.

4.7.3.1.5 3A6 TEST switch S5. Depress 3A6 TEST switch S5 three times and verify that 3A6 TEST indicator DS5 is alternately illuminated and extinguished.

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4.7.3.1.6 Combinations of switches S1, S2, S3, S4 and S5. Turn IF AMPLITUDE ADJ control AT4 to the zero (00) position.

a. Connect test equipment as in Figure 8. Remove 50 ohm termination from test harness terminal TB3-J1-D and attached coaxial lead Z to this terminal. For each of the switchcombinations A, D, E, F, K, L, M, P of Table VI, verify that the width, amplitude and delay of the bursts on channel B conform to the requirements of IF INPUT UNIT 3, Table V. Verify for switch combinations F, L, M, P that 59.936 MHz excursions on B channel are synchronous (successive recurrences superimposed). Verify for switch combinations A, D, E, K that 59.941 MHz excursions on channel B are asynchronous. Remove coaxial lead Z from TB3-J1-D and restore 50 ohm termination.

b. Remove 50 ohm termination from Receiver Test Panel terminal J11 and attach coaxial lead Z to this terminal. For each of the switch combinations A, D, E, F, K, L, M, P of Table VI, verify that the width, amplitude and delay of the bursts on channel B conform to the requirements of IF OUT, Table V. Verify for switch combinations F, L, M, P that 59.936 MHz excursions on B channel are synchronous (successive recurrences superimposed). Verify for switch combinations A, D, E, K that 59.941 MHz excursions on channel B are asynchronous and the 59.936 MHz burst

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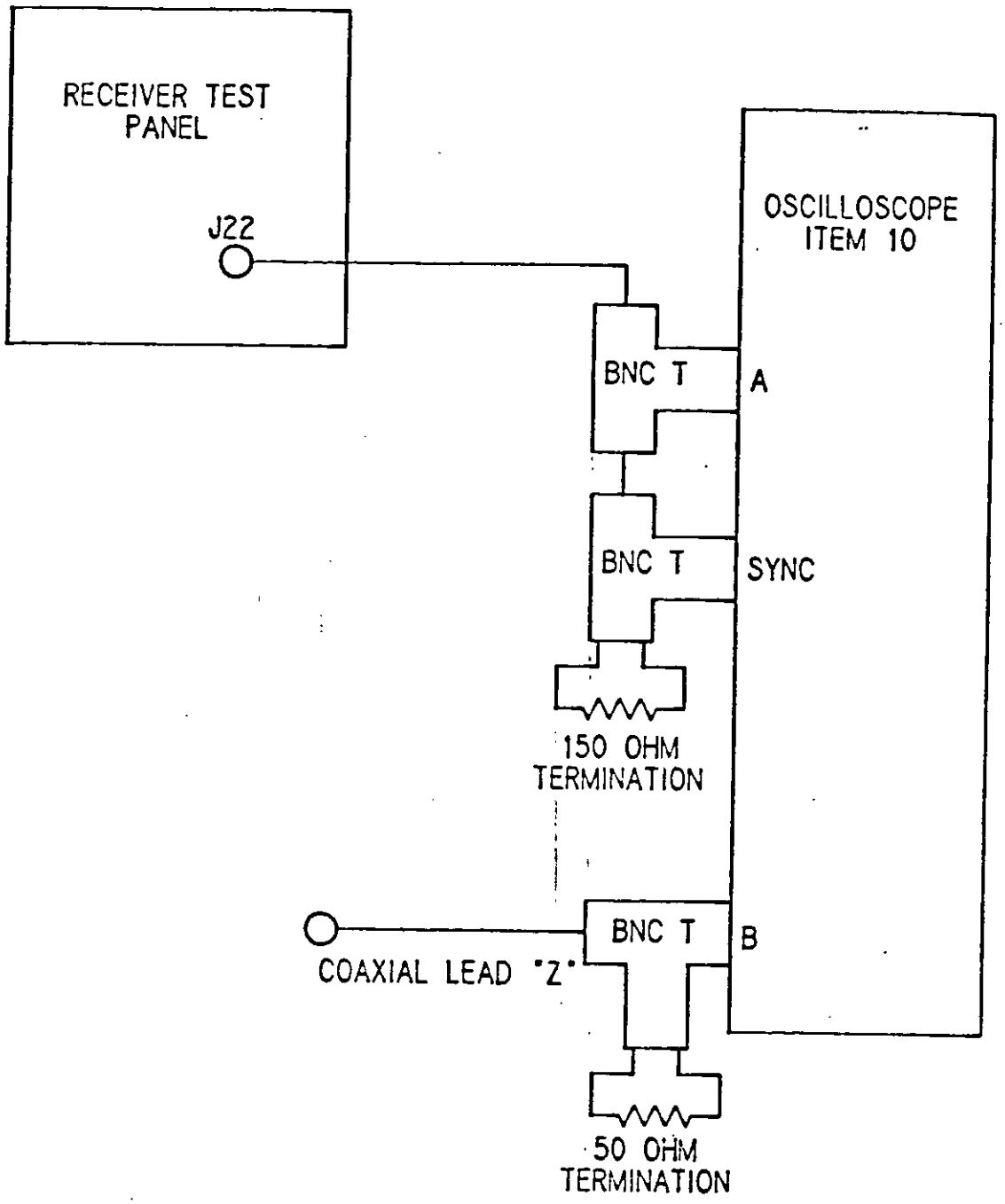


FIGURE 8. GATED STALO OSC check #1. RANGE CAL RF INPUT Check combination of switches S1, S2, S3, S4 and S5 checks.

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is superimposed on the 59.941 continuous wave at Combination Indicators C, H, J, O. Remove coaxial lead Z from J11 and restore 50 ohm termination.

c. Remove 50 ohm termination from Receiver Test Panel terminal J12 and attach coaxial lead Z to this terminal. For each of the switch combinations A, D, E, F, K, L, M, P of Table VI, verify that the width, amplitude and delay of the bursts on channel B conform to the requirement of IF OUT 3A6, Table V. Verify for switch combinations F, L, M, P that 59.936 MHz excursions on B channel are synchronous (successive recurrences superimposed). Verify for switch combinations A, D, E, K that 59.941 MHz excursions on channel B are asynchronous. Remove coaxial lead Z from J12 and restore 50 ohm termination.

d. Inside the Receiver Test Panel, open the connection between P60 and J33, at the output of test oscillator, 10554934 and insert a TPS T connector. Attach TPS-to-BNC adapter to the third port of the connector, and complete connections to set up test equipment according to Figure 9. Remove 50 ohm termination from test harness terminal TB3-J1-D and attach coaxial lead Z to this terminal. Set up switch combinations A, D, E, K of Table VI. Verify that 59.941 MHz excursions on both channels are asynchronous, but that envelope of burst on B channel is stable. Set up switch combination C, H, J, O of Table VI. Verify that 59.936 MHz burst superimposes the 59.941 MHz continuous wave. Disconnect P60 from the TPS T connector in receiver test panel and verify that signal on B channel only disappears. Reconnect P60 to the TPS T connector. Restore 50 ohm termination at TB3-J1-D.

e. Remove 50 ohm termination from Receiver Test Panel terminal J12 and attach coaxial lead Z to this terminal. Set up switch combinations A, D, E, K of Table VI. Verify that 59.941 MHz excursions on both channels are asynchronous, but that envelope of bursts on B channel is stable. Disconnect P60 from the TPS T connector in the receiver test panel and verify that signal on B channel only disappears. Reconnect P60 to the TPS T connector. Remove coaxial lead Z from J12.

f. Remove coaxial lead Y from TPS to BNC adapter (see Figure 9). Attach electronic counter (item 12, Table XIII) to adapter. Verify that frequency of signal is 59.941 $\pm$ 0.001 MHz.

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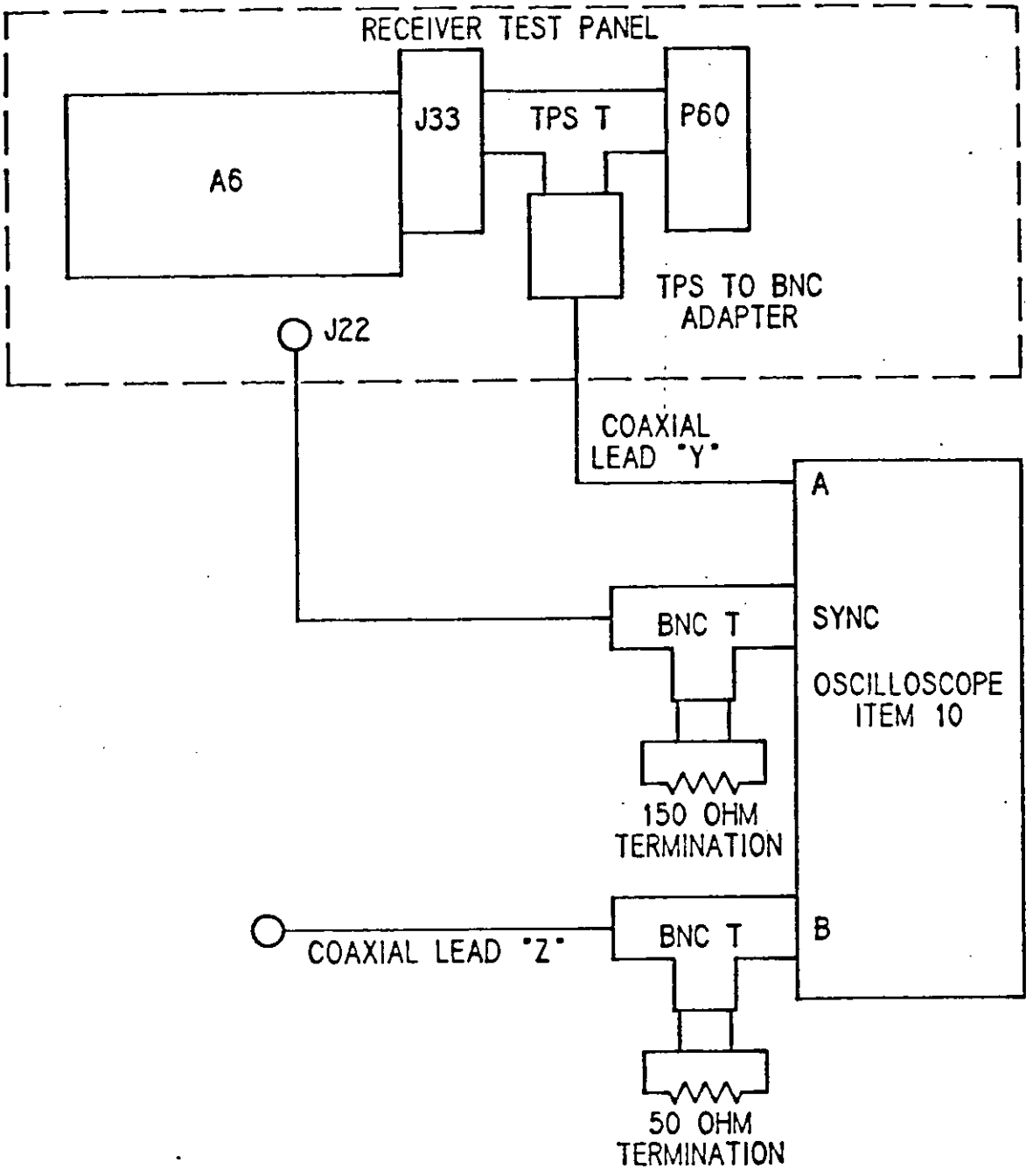


FIGURE 9. 59.941 MHz burst check.

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4.7.3.2 SYNC TRIGGER CONTROL. Reestablish conditions existing before MODULATOR TRIGGER check of 4.7.3.1.1. Turn switch S1 of test harness, Figure 1, to position 2. Connect test equipment as in Figure 4. Perform the following steps in order.

a. MODULATOR TRIGGER. Verify that MODULATOR TRIGGER signal of Table II appears on oscilloscope.

b. GATED STALO OUTPUT. Connect test equipment as in Figure 8. Remove 50 ohm termination from Receiver Test Panel terminal J25 and attach coaxial lead Z to this terminal. Adjust oscilloscope to display approximately 10 cycles on A channel. Verify that bursts on B channel are constant and occur in a one-for-one relationship to the pulses on channel A. Verify that the amplitude, width and delay of bursts on channel B conform to the requirements of GATED STALO OUTPUT, Table VII. Connect test equipment as in Figure 10. Expand display so as to spread one burst on channel B across the face of the oscilloscope. Superimpose A and B channel displays and verify that 59.936 MHz excursions on the two channels occur in a one for one relationship.

c. STALO OSC UNIT II. Remove 50 ohm termination from Receiver Test Panel J23. Connect test equipment as in Figure 11. Verify that amplitude of signal is constant and conforms to requirements of STALO OSC UNIT II, Table VII. Reconnect test equipment as in Figure 5, except with coaxial lead connected to test harness terminal J23. Verify that frequency of signal conforms to requirement of STALO OSC UNIT II, Table VII. Remove coaxial lead from J23 and restore 50 ohm termination at that point.

4.7.3.3 IF DELAY ADJ controls.

a. Reestablish conditions existing at the end of the fifth sentence of 4.7.3.2b. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.3.

b. Reestablish conditions existing at the end of the third sentence of 4.7.3.1.3b. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.3.

c. Reestablish conditions existing at the end of the third sentence of 4.7.3.1.6a. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.3.

d. Reestablish conditions existing at the end of the second sentence of 4.7.3.1.6b. Verify that IF DELAY ADJ controls

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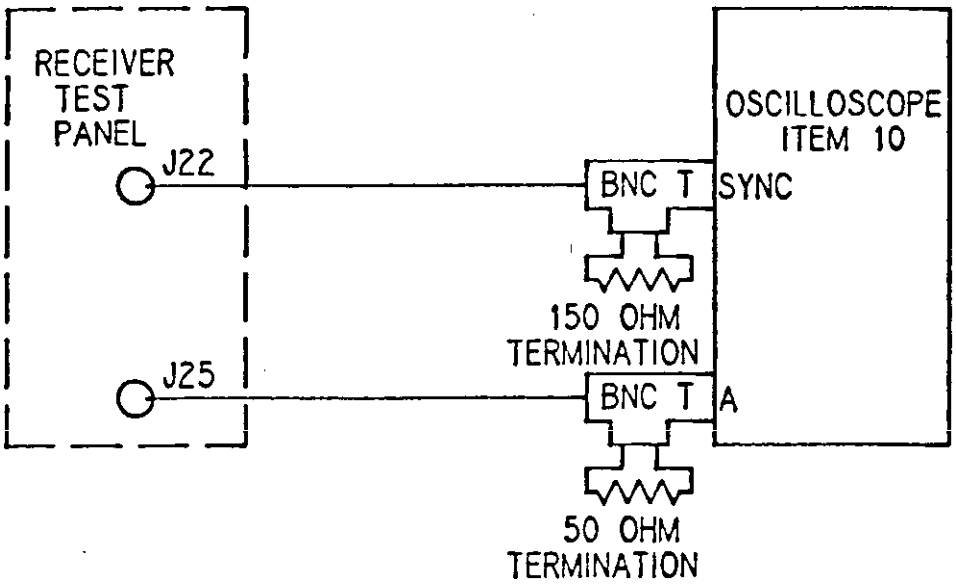


FIGURE 10. GATED STALO OSC check #1.

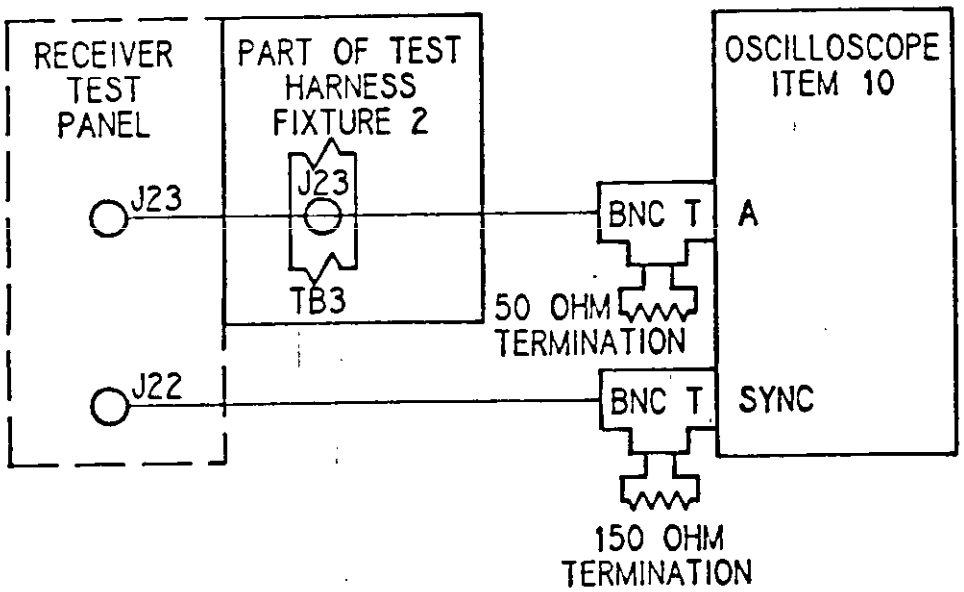


FIGURE 11. STALO OSC unit II check.

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affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.3.

e. Reestablish conditions existing at the end of the second sentence of 4.7.3.1.6c. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.3.

f. Reestablish conditions existing at the end of the fifth sentence of 4.7.3.1.1e. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.3.

4.7.3.4 IF AMPLITUDE ADJ control.

a. Reestablish conditions existing at the end of the third sentence of 4.7.3.1.6a. Verify that IF AMPLITUDE ADJ control affects the amplitude of signal on B channel of oscilloscope in accordance with the requirements of 3.5.4.

b. Reestablsh conditions existing at the end of the second sentence of 4.7.3.1.6b. Verify that IF AMPLITUDE ADJ control affects the amplitude signal on B channel of oscilloscope in accordance with the requirements of 3.5.4.

c. Reestablish conditions existing at the end of the second sentence of 4.7.3.1.6c. Verify that IF AMPLITUDE ADJ control affects the amplitude of signal on B channel of oscilloscope in accordance with the requirements of 3.5.4.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS10554527.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Receiver Test Panel has two modes of operation. In the SYNC TRIGGER CONTROL mode, the Receiver Test Panel serves simply as an adjunct to Test Set, Subassembly MX-8046, the Receiver/Transmitter Test Panel. The latter panel is used in testing Unit 2, the Receiver/Transmitter, of Radar Set AN/VPS-2. In this mode the Receiver Test Panel supplies certain voltages and signals, and only two of its front panel controls have any effect on its performance.

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6.2 Acquisition Requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certification of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire control system
Radar Set, AN/VPS-2
Radar Test Set
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A305)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-50333C (AR)	4 DECEMBER 1991

3. DOCUMENT TITLE: TEST SET, SUBASSEMBLY MX-8048/TPM-22A (RECEIVER, TEST PANEL)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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